Operational Labels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Operational Labels before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a **label**led second line to correct your explanation.

Rows Columns and Data Types

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Loops and Stopping

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Decompose a Goal

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Operational Labels trace and record the first step you can explain without help.

Use W2: Replace a vague instruction for Categories and operational **label**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Operational Labels explanation against the supplied facts before discussing it.

Use the worked case for Categories and operational **label**s. Proposed explanation: “A blank is the same as a failing value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Two **label**lers use book-length rules: fewer than 120 pages and at most 120 pages. **Label** books with 110,120,121 pages. The agreed policy is at most 120. Repair the **disagreement**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Operational Labels target to a trace and a justified conclusion.

Two **label**ers use 'short' to mean pages<100 and pages<=100. They disagree only on 100-page books. Write a shared definition and explain how to repair the existing **label**s.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Operational Labels case and show the deciding evidence.

Short means fewer than 100 pages. Two **annotator**s disagree about a 100-page book.

Resolve the **label**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Use a clear **label** rule: short means 80 pages or fewer. Two judges **label** an 80-page book differently. What shared instruction resolves this **disagreement**?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 8 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Operational Labels adds to Transparent Book Recommender and keep the evidence.

Define ‘good match’ before **training** or evaluating the recommender.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Two **label**ers use 'short' to mean pages<100 and pages<=100. They disagree only on 100-page books. Write a shared definition and explain how to repair the existing **label**s.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Apply the calendar guide

> Apply the calendar guide. Apply the supplied Operational Labels facts and keep your reasoning.

The calendar policy accepts whole-number days 1 through 30 inclusive and a room code A or B. Check known failures first, then unknown fields, then readiness.

Label these cards and give **evidence**: day 30 room B; day 31 room A; day 14 room unreadable; day 31 room unreadable.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Replace a vague instruction

> Replace a vague instruction. Apply the supplied Operational Labels facts and keep your reasoning.

A helper says **Label** a card ready if it looks nice. Explain why this can lead to inconsistent **label**s.

Replace it with an instruction that another person can apply using the calendar policy, including what to do about a blank field.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Count and investigate disagreement

> Count and investigate disagreement. Apply the supplied Operational Labels facts and keep your reasoning.

For the same four cards, Mina records ready, revise, review, ready. Theo records ready, review, review, revise. How many positions agree? Identify the **disagreement**s.

Can you decide which **annotator** is correct from these **label** lists alone? Explain what else you need.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Transfer the method to a new record

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the method to a new record. Apply the

> supplied Operational Labels facts and keep your reasoning.

A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y.

Label revise if any known field fails; otherwise **label** review if a field is blank or unreadable; otherwise **label** ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code.

Explain whether ready alone grants permission to borrow.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Keep a policy change visible

> Keep a policy change visible. Apply the supplied Operational Labels facts and keep your reasoning.

Version 1 accepts calendar days 1 through 30. Version 2 accepts days 1 through 31. A card says day 31 room A. State its **label** under each version.

Explain why mixing the two **label**s without recording the version could create a misleading **training** dataset.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Operational Labels

> Two fictional catalogue reviewers label books compact or extended. Explain the labelled step

> before checking the changed condition.

Two fictional catalogue reviewers **label** books compact or extended.

Agreed policy: compact means at most 80 pages; extended means more than 80.

Reviewer A uses pages<80; reviewer B uses pages<=80. Inspect books of 79,80,81 pages.

AI ACADEMY / AI-2/8

Operational label boundary

	Pages	A labels	B labels
Book 1	79	compact	compact
Book 2	80	extended	compact
Book 3	81	extended	extended

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/8

Operational Labels

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.



1. Shared definition

At most 80 includes a book with exactly 80 pages.



2. Reviewer A

79 compact; 80 and 81 extended.



3. Reviewer B

79 and 80 compact; 81 extended.



4. Repair boundary

A must use <=80; retain the disagreement and changed policy implementation.

CHANGE ONE THING
Change the documented policy to fewer than 80 instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why can we not resolve the **disagreement** merely by taking a majority of reviewers?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Give Transparent Book Recommender metadata **label**s explicit definitions and boundary examples for reviewers.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Give Transparent Book Recommender metadata **label**s explicit definitions and boundary examples for reviewers.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two fictional catalogue reviewers **label** books compact or extended.

Practice 1: explain the deciding evidence

Which **label**led step matches this detail? A must use <=80; retain the **disagreement** and changed policy implementation.

1. Repair boundary

2. Reviewer B

3. Reviewer A

4. Shared definition

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair boundary A must use <=80; retain the **disagreement** and changed policy implementation.

The reviewers agree on two of three examples, but that fraction does not decide which definition to keep. The documented policy decides the equality case. Repairing A's implementation aligns its **label**s with the agreed meaning.

Rechecking 79 and 81 ensures the boundary fix preserves adjacent decisions.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the documented policy to fewer than 80 instead. Which conclusion is supported by the supplied facts?

1. A's original **label**s now match; B must change its equality behavior.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A's original **label**s now match; B must change its equality behavior. The authoritative definition changed, not the physical books.

Repair a missing building block

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?.

Return to this week and explain what you changed.

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

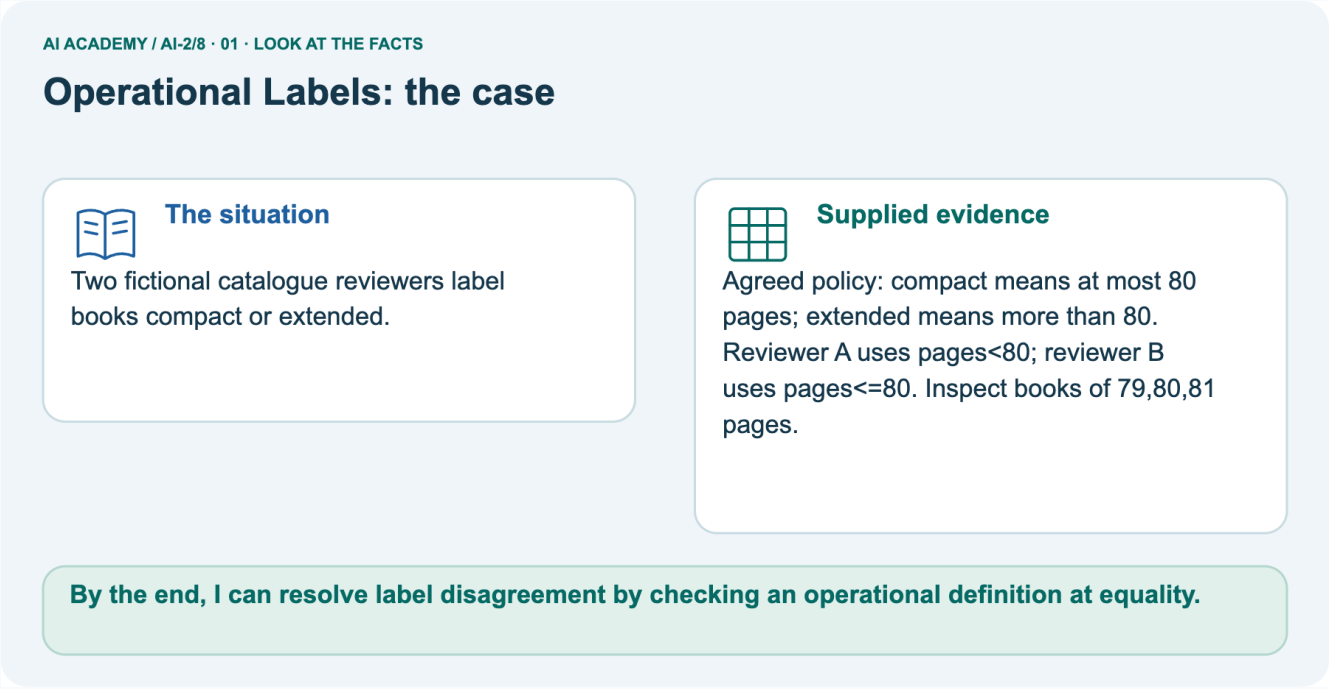
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Operational Labels picture from its labels, then record a relevant limit.

01. Operational Labels: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

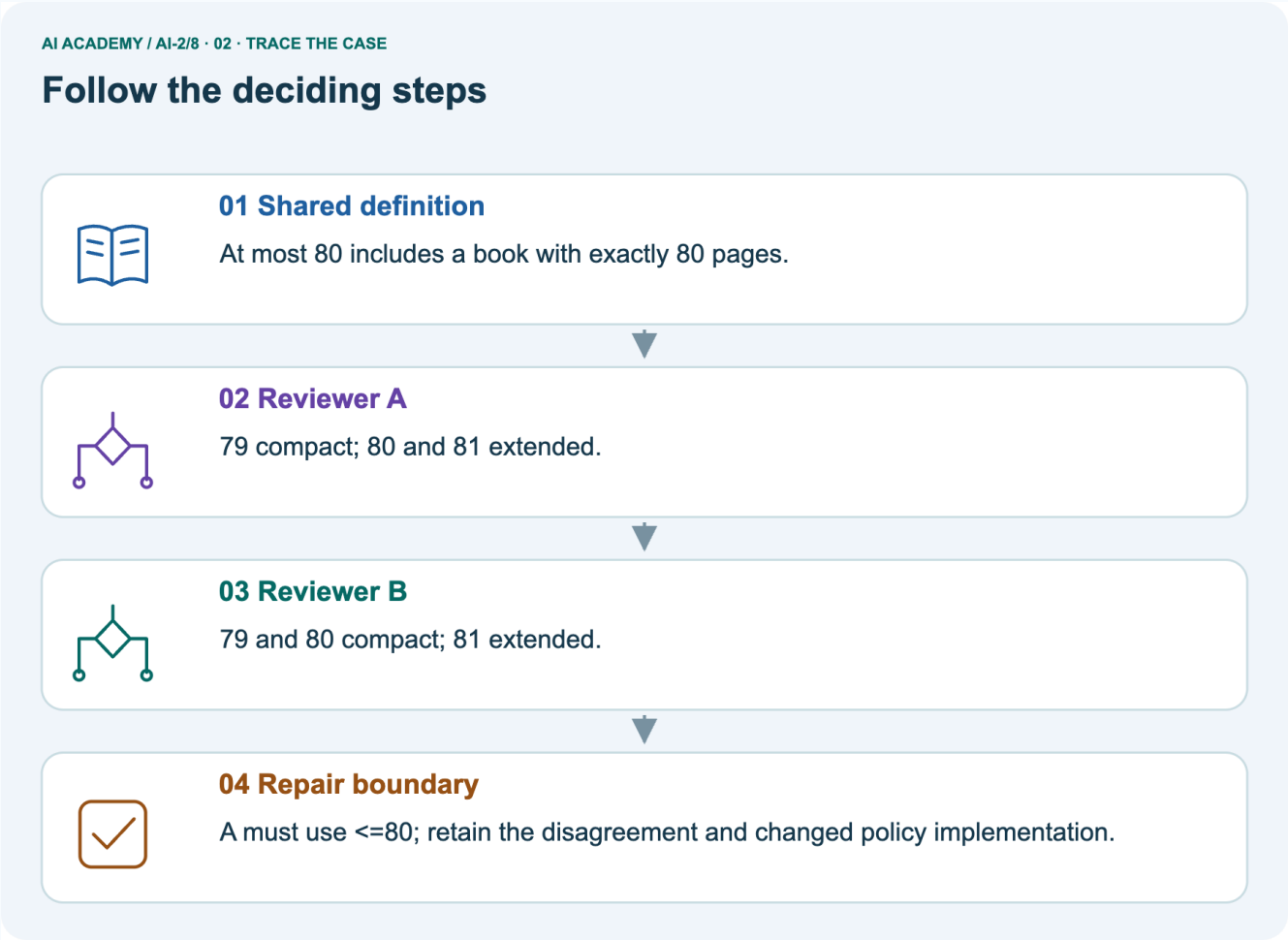
Two fictional catalogue reviewers **label** books compact or extended. Supplied facts: Agreed policy: compact means at most 80 pages; extended means more than 80. Reviewer A uses pages<80; reviewer B uses pages<=80.

Inspect books of 79,80,81 pages.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



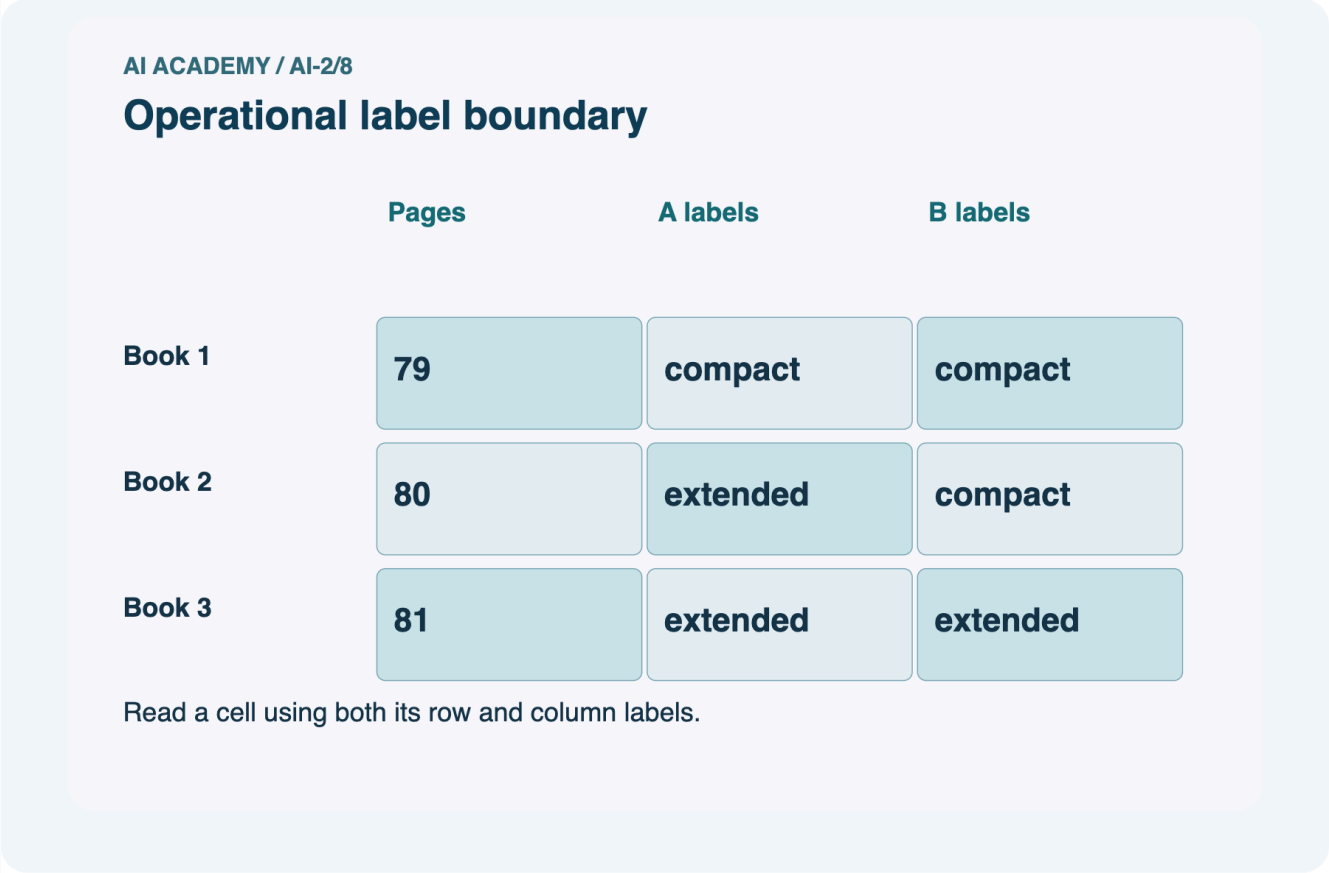
Follow the labelled picture with the matching explanation. Open the vector diagram

Shared definition: At most 80 includes a book with exactly 80 pages. → Reviewer A: 79 compact; 80 and 81 extended. → Reviewer B: 79 and 80 compact; 81 extended. → Repair boundary: A must use <=80; retain the **disagreement** and changed policy implementation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Operational label boundary



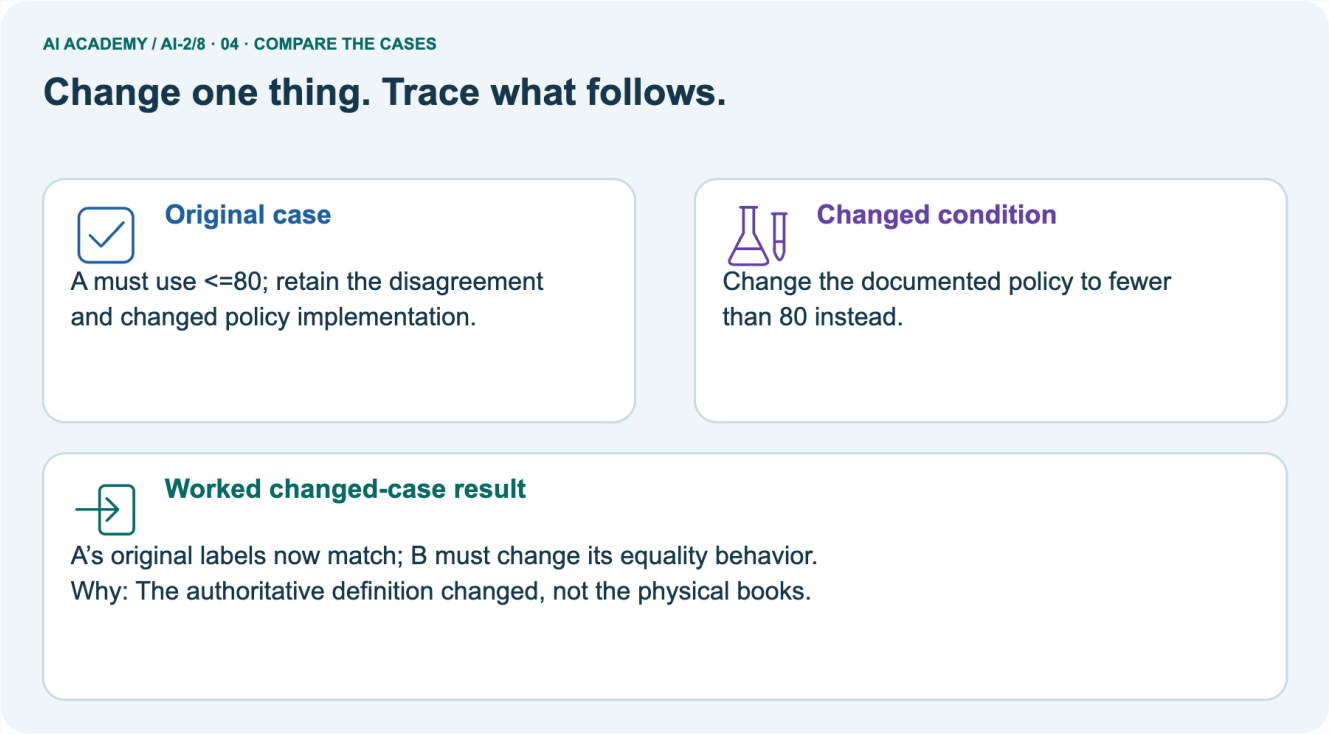
Follow the labelled picture with the matching explanation. Open the vector diagram

Operational **label** boundary. Book 1: Pages = 79, A **label**s = compact, B **label**s = compact. Book 2: Pages = 80, A **label**s = extended, B **label**s = compact. Book 3: Pages = 81, A **label**s = extended, B **label**s = extended.

Read the **label**s and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

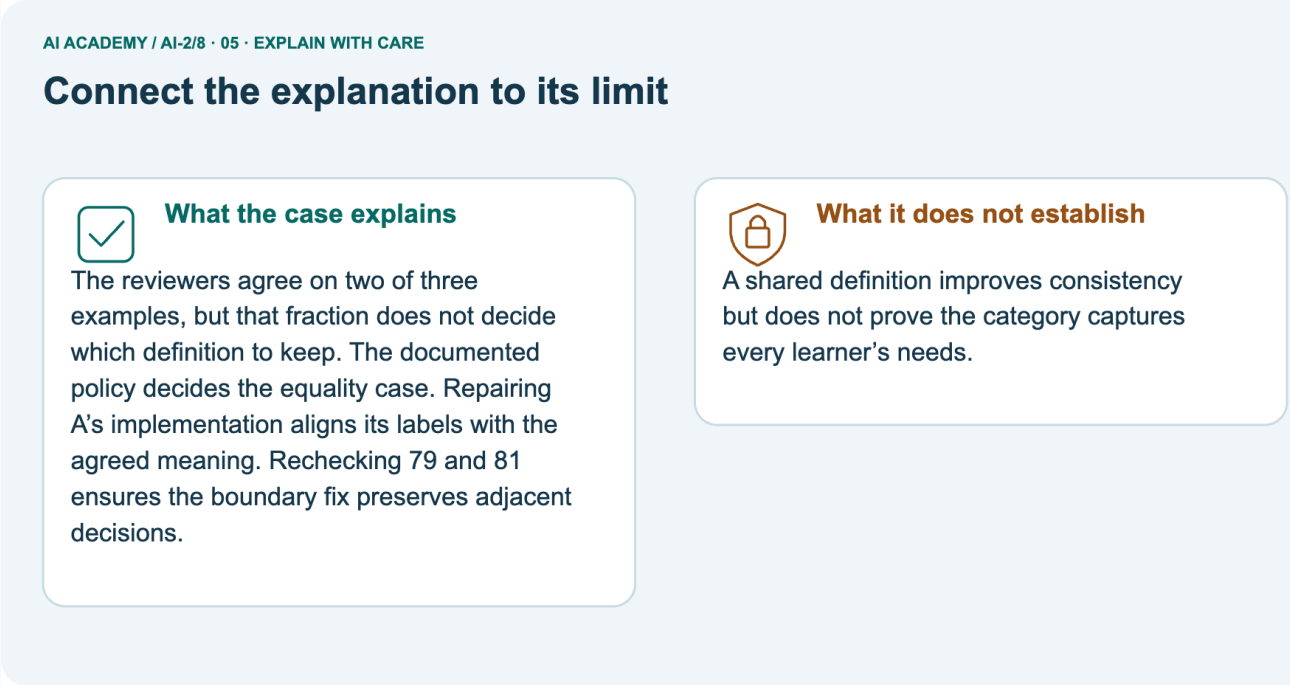
Original case: A must use <=80; retain the **disagreement** and changed policy implementation.
Changed condition: Change the documented policy to fewer than 80 instead.

Expected result: A's original **label**s now match; B must change its equality behavior. Reason: The authoritative definition changed, not the physical books.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

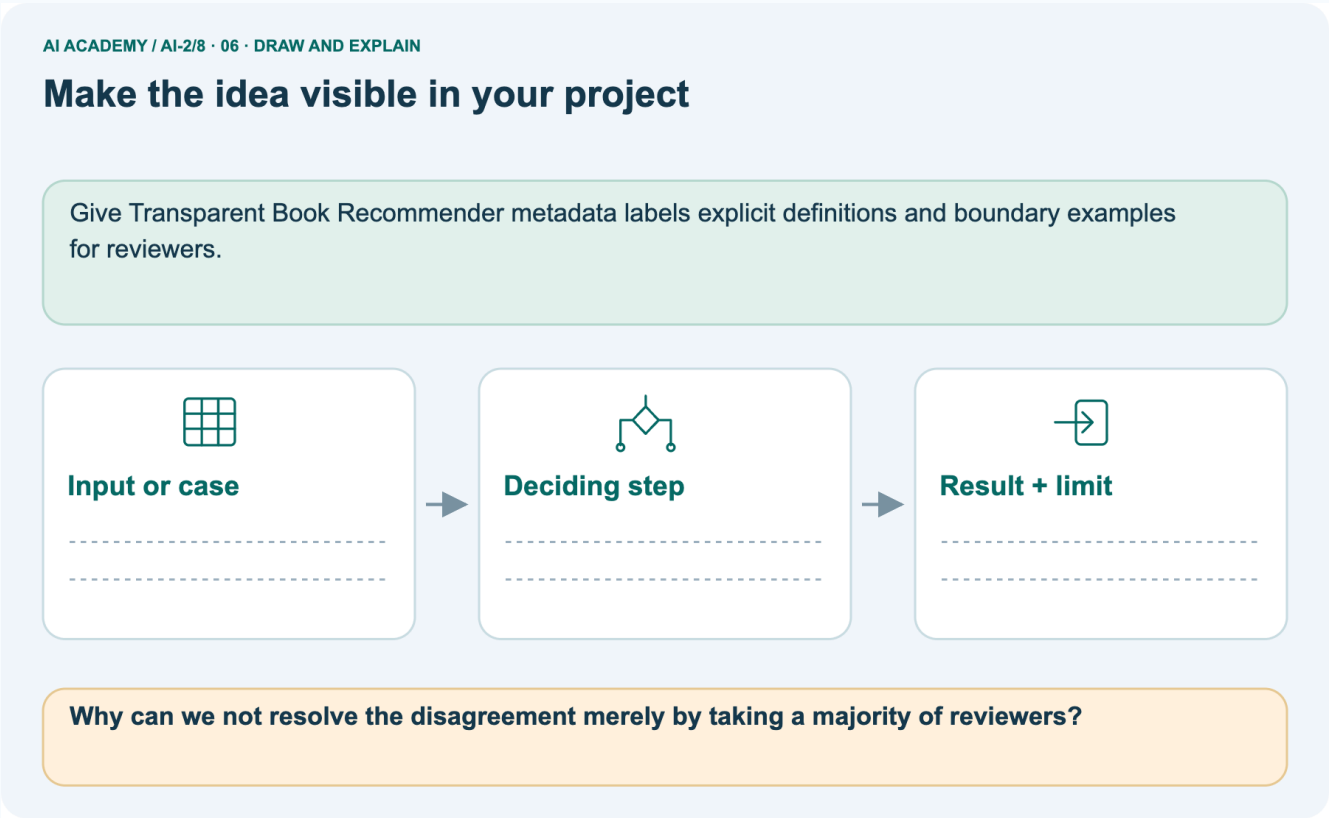
The reviewers agree on two of three examples, but that fraction does not decide which definition to keep. The documented policy decides the equality case. Repairing A's implementation aligns its **label**s with the agreed meaning.

Rechecking 79 and 81 ensures the boundary fix preserves adjacent decisions. Limit: A shared definition improves consistency but does not prove the **category** captures every learner's needs.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give Transparent Book Recommender metadata **label**s explicit definitions and boundary examples for reviewers. Draw your **input**, deciding step, result and limit.

Supported question: Why can we not resolve the **disagreement** merely by taking a majority of reviewers?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.